

AQI ANALYSIS

using R

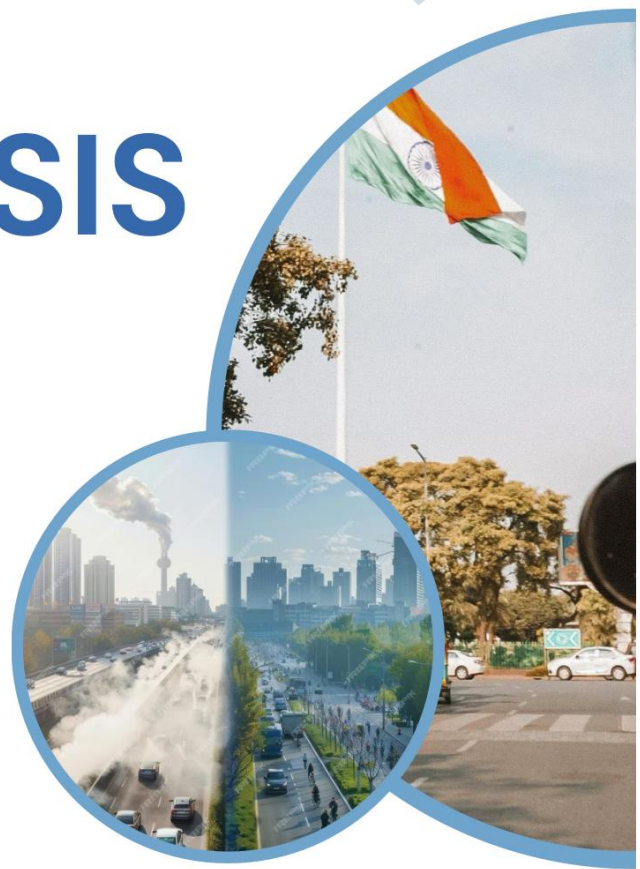
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INDEX

Sr No.	Topic	Page Number
1.	Introduction	3
2.	Why was the data chosen?	4
3.	Summary statistics	4
4.	Data cleaning	5
5.	Choosing of the statistical methods used	5
6.	Overview of the statistical methods used	6
7.	Charts and visualisations	6
8.	Inferences	14
9.	Final Conclusion	18

INTRODUCTION

This report digs into air quality data from states and cities across India, using R to run stats and make some visuals. With all the urban sprawl and factory growth here, pollution's turned into a real headache for public health and the environment.

We measure air quality through stuff like CO, NO₂, NH₃, SO₂, ozone, PM10, and especially PM2.5—the fine particles that really mess with your lungs and heart. Looking at concentrations by state and city helps show where pollution's worst and how it's spread out.

Dated February 10, 2026, the study pulls real-time data from December 30, 2025, to spot patterns, stack regions against each other, and link pollutants like PM2.5, PM10, NO₂, SO₂, CO, NH₃, and ozone. Recent news shows the mess: almost half the cities tracked blew past PM2.5 limits in January 2026, with places like Ghaziabad and Delhi leading the pack.

India's got this ongoing pollution problem of 44% of cities stuck with high PM2.5 from cars, factories, and burning crop waste. We tackled it with a dataset of 2,520 observations, covering latitude longitude from 8.5–34.1 and 70.9–94.6 across stations. Delhi tops the state averages at 134 (followed by Himachal, Chandigarh, Rajasthan, and Odisha), way over the national 65.17, which really highlights the gaps between regions.

Hypothesis testing on PM2.5 vs PM10 came out striking. PM2.5 levels are significantly higher on average, and that's worrying since those tiny particles dig deeper into your lungs. The ANOVA was wild too, with a p-value basically at zero, meaning pollution levels vary a ton across different pollutant types.

The simple linear regression (SLR) had R² of 0.82 which shows minimum pollution levels explain a good chunk of the averages.

Why was the Data chosen?

[Dataset Used](#) (click to view)

The dataset was selected because air pollution is both a practical and statistically meaningful issue. It has direct implications for public health and economic activity, and it also provides structured numerical data suitable for statistical analysis.

The presence of multiple pollutant categories such as PM2.5, PM10, CO, NO2, NH3, Ozone, and SO2 allows for meaningful comparison across groups. Including latitude and longitude makes it possible to examine spatial variation through regression analysis. In addition, having minimum, maximum, and average pollutant values makes it easier to study dispersion and distribution patterns.

Another important feature is the coverage across several states and cities, which supports regional comparison and ranking. From a methodological perspective, the dataset allows the use of descriptive statistics, hypothesis testing, ANOVA, and regression within the same framework. It is detailed enough to produce meaningful results without being overly complex.

Summary statistics

The summary statistics show that pollution levels are not evenly distributed. The mean of pollutant_avg is approximately 65.17, while the median is 39. The difference between these two measures suggests positive skewness, meaning that a smaller number of high values are increasing the overall average.

The standard deviation, about 75.26, indicates considerable variability across monitoring stations. Pollution values range from 1 to 449, which reflects a wide spread in intensity. Even within the middle 50% of the data, the interquartile range (IQR) of 76 shows noticeable dispersion.

Pollutant categories are fairly balanced in terms of frequency, which makes comparisons across types more reliable. Overall, the descriptive statistics suggest that pollution exposure varies significantly across locations, and the skewed distribution supports the need for further inferential analysis rather than relying only on averages.

Data Cleaning

Before beginning the analysis, the dataset was prepared to ensure consistency and compatibility with statistical software. The original file was renamed for clarity and converted from CSV format to XLSX to make handling and organization easier. During this process, all character values representing numerical information were converted into numeric format so that they could be used correctly in calculations and modeling. The date column was also assigned the appropriate date data type to ensure accurate interpretation and avoid formatting issues during analysis.

These initial steps were important for preventing type-related errors in statistical procedures. Once the data types were corrected, the dataset was reviewed to confirm that numerical variables were properly recognized and that no unintended text formatting remained. This preparation ensured that the dataset was clean, structured, and ready for reliable statistical analysis.

STATISTICAL METHODS

Choosing of the Statistical Methods Used

The statistical methods were selected based on the research questions and the nature of the variables. Descriptive statistics were used first to understand the basic distribution and variability in pollution levels.

For comparing PM_{2.5} and PM₁₀, a two-sample t-test was chosen because it compares independent group means and does not assume equal variances. Since the dataset contains more than two pollutant categories, one-way ANOVA was appropriate for testing overall differences across groups.

Simple linear regression was used to examine the relationship between pollutant_min and pollutant_avg, as both variables are continuous. Multiple linear regression was then applied to assess the combined effect of latitude and longitude on pollution levels.

Each method was selected for its relevance to the specific analytical objective. Together, these techniques allow both comparison and relationship analysis within a consistent statistical framework.

Overview of the Statistical Methods Used

The analysis combines descriptive and inferential statistical methods. Descriptive statistics summarize the distribution of pollution levels through measures such as mean, median, standard deviation, and range.

The two-sample t-test compares average pollution between PM2.5 and PM10 and shows a statistically significant difference. One-way ANOVA tests whether pollution levels differ across all pollutant categories, and the significant F-statistic indicates variation among groups.

Simple linear regression measures the association between pollutant_min and pollutant_avg. The R^2 value of approximately 0.817 suggests a strong relationship. Multiple linear regression examines the impact of latitude and longitude; although statistically significant, the R^2 of about 0.033 indicates limited explanatory strength.

Overall, these methods were applied to understand both distributional characteristics and structural relationships within the dataset, rather than relying on isolated summary measures.

CHARTS AND VISUALIZATIONS

1. Dataset Structure and Summary

```
str(air)

## tibble [2,520 × 11] (S3: tbl_df/tbl/data.frame)
## $ country      : chr [1:2520] "India" "India" "India" "India" ...
## $ state        : chr [1:2520] "Assam" "Assam" "Assam" "Assam" ...
## $ city         : chr [1:2520] "Guwahati" "Guwahati" "Guwahati" "Guwahati" ...
## $ station      : chr [1:2520] "LGBI Airport, Guwahati - PCBA" "LGBI
LGBI Airport, Guwahati - PCBA" "LGBI Airport, Guwahati - PCBA" "Railway Colony,
Guwahati - PCBA" ...
## $ last_update  : POSIXct[1:2520], format: "2025-12-30 21:00:00" "2025-12-
2025-12-30 21:00:00" ...
## $ latitude     : num [1:2520] 26.1 26.1 26.1 26.2 26.3 ...
## $ longitude    : num [1:2520] 91.6 91.6 91.6 91.8 92.7 ...
## $ pollutant_id : chr [1:2520] "NO2" "NH3" "SO2" "PM2.5" ...
## $ pollutant_min: num [1:2520] 13 5 2 2 125 1 4 9 93 19 ...
## $ pollutant_max: num [1:2520] 210 22 35 2 153 1 4 9 117 24 ...
## $ pollutant_avg: num [1:2520] 23 6 14 2 137 1 4 9 101 22 ...

summary(air)
```

```
##      country          state          city          station
## Length:2520      Length:2520      Length:2520      Length:2520
## Class :character  Class :character  Class :character  Class :character
## Mode  :character  Mode  :character  Mode  :character  Mode  :character
##
##
##      last_update          latitude          longitude
## Min.      :2025-12-30 21:00:00      Min.      : 8.515      Min.      :70.91
## 1st Qu.:2025-12-30 21:00:00      1st Qu.:19.063      1st Qu.:74.73
## Median :2025-12-30 21:00:00      Median :23.108      Median :77.11
## Mean      :2025-12-30 21:00:00      Mean      :22.645      Mean      :78.05
## 3rd Qu.:2025-12-30 21:00:00      3rd Qu.:26.950      3rd Qu.:79.96
## Max.      :2025-12-30 21:00:00      Max.      :34.066      Max.      :94.64
## pollutant_id      pollutant_min      pollutant_max      pollutant_avg
## Length:2520      Min.      : 1.0      Min.      : 1.0      Min.      : 1.00
## Class :character  1st Qu.: 6.0      1st Qu.: 19.0      1st Qu.: 14.00
## Mode  :character  Median : 18.0      Median : 66.0      Median : 39.00
##                      Mean      : 36.9      Mean      :105.4      Mean      : 65.17
##                      3rd Qu.: 48.0      3rd Qu.:137.0      3rd Qu.: 90.00
##                      Max.      :366.0      Max.      :500.0      Max.      :449.00
```

2. Descriptive Statistics

```
mean(air$pollutant_avg)
## [1] 65.16865
median(air$pollutant_avg)
## [1] 39
sd(air$pollutant_avg)
## [1] 75.25719
min(air$pollutant_avg)
## [1] 1
max(air$pollutant_avg)
## [1] 449
IQR(air$pollutant_avg)
## [1] 76
```

3. Categorical Distributions

```
table(air$pollutant_id)

##
##      CO      NH3      NO2 OZONE      PM10      PM2.5      SO2
##    385     320     363    380     357     363     352

prop.table(table(air$pollutant_id))

##
##           CO           NH3           NO2           OZONE           PM10           PM2.5           SO2
## 0.1527778 0.1269841 0.1440476 0.1507937 0.1416667 0.1440476 0.1396825
```

4. State-wise Ranking of Pollution

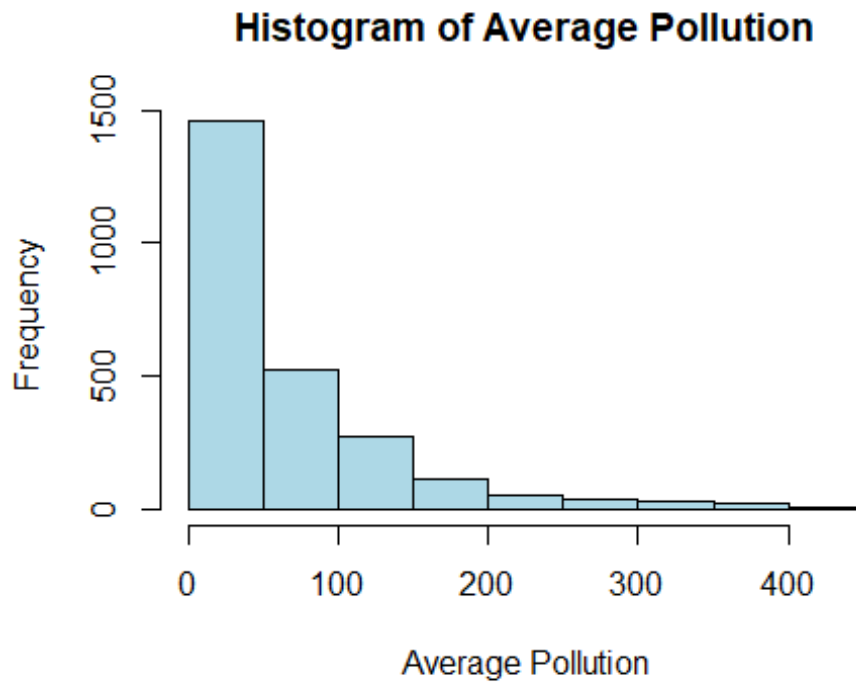
```
state_rank <- air %>%
  group_by(state) %>%
  summarise(
    avg_pollution = mean(pollutant_avg),
    observations = n()
  ) %>%
  arrange(desc(avg_pollution))

head(state_rank, 10)

## # A tibble: 10 × 3
##   state          avg_pollution observations
##   <chr>          <dbl>          <int>
## 1 Delhi          134.             247
## 2 Himachal Pradesh 111.             7
## 3 Chandigarh      85.5            15
## 4 Odisha          84.8            116
## 5 Jharkhand       74.3             6
## 6 Puducherry       68              7
## 7 Rajasthan       67.6            313
## 8 Haryana         67.5            192
## 9 Maharashtra     62.2            512
## 10 Gujarat        61.1             92
```

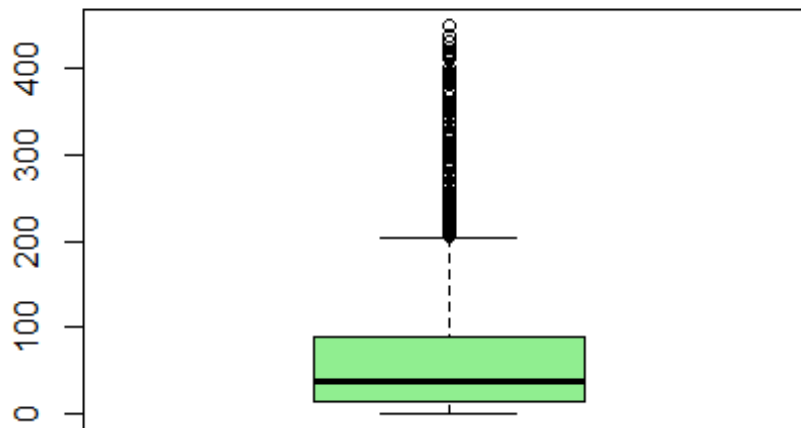

5. Data visualization

```
hist(air$pollutant_avg,  
     main = "Histogram of Average Pollution",  
     xlab = "Average Pollution",  
     col = "lightblue")
```

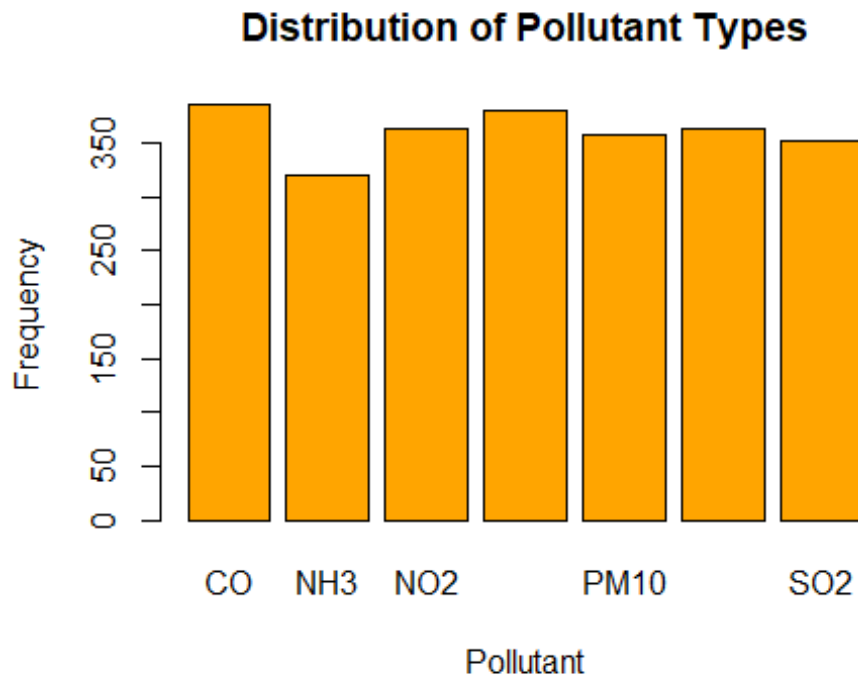


```
boxplot(air$pollutant_avg,  
        main = "Boxplot of Average Pollution",  
        col = "lightgreen")
```

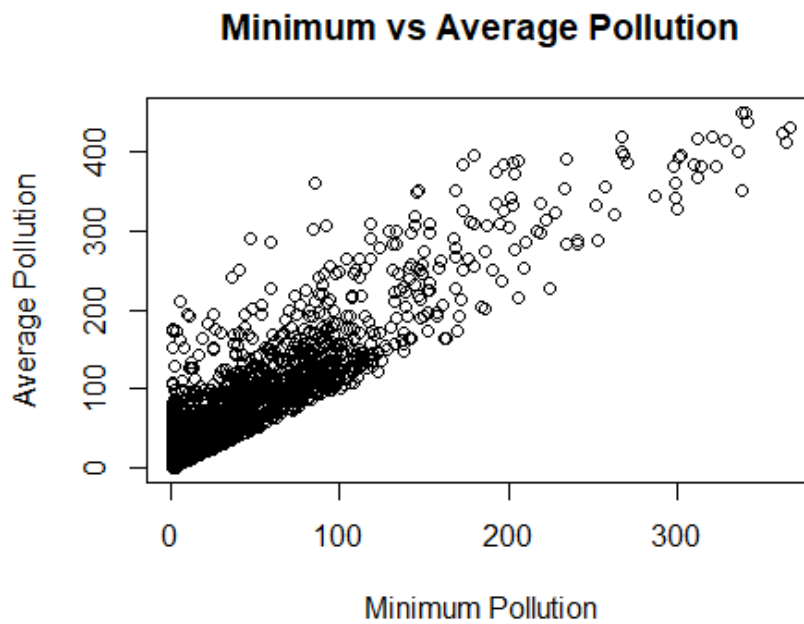
Boxplot of Average Pollution



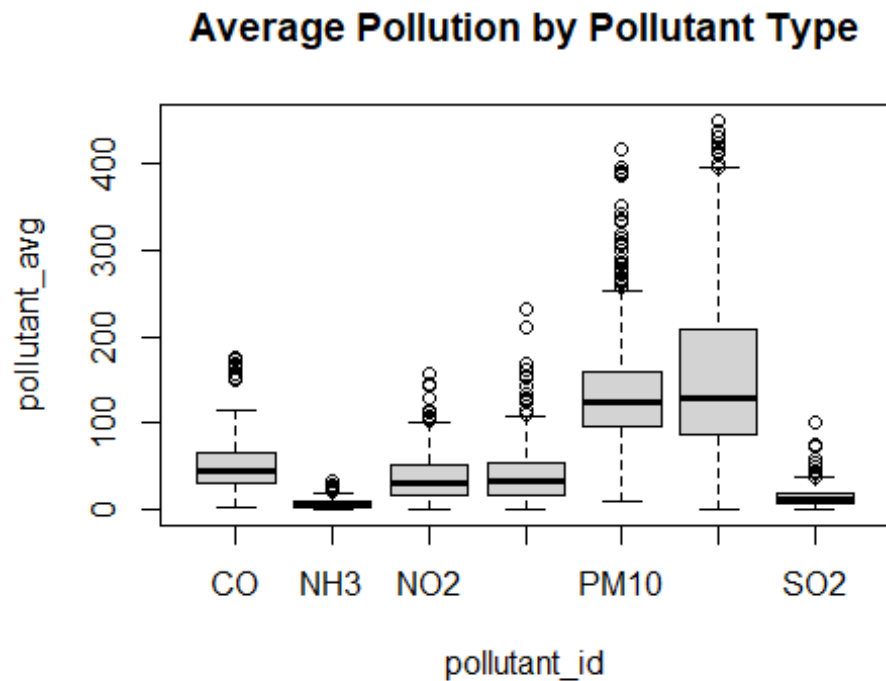
```
barplot(table(air$pollutant_id),  
  main = "Distribution of Pollutant Types",  
  xlab = "Pollutant",  
  ylab = "Frequency",  
  col = "orange")
```



```
plot(air$pollutant_min, air$pollutant_avg,  
     main = "Minimum vs Average Pollution",  
     xlab = "Minimum Pollution",  
     ylab = "Average Pollution")
```



```
boxplot(pollutant_avg ~ pollutant_id,
        data = air,
        main = "Average Pollution by Pollutant Type")
```



6. Hypothesis Testing

```
pm_data <- subset(air, pollutant_id %in% c("PM2.5", "PM10"))
t.test(pollutant_avg ~ pollutant_id, data = pm_data)

##
## Welch Two Sample t-test
##
## data: pollutant_avg by pollutant_id
## t = -3.1284, df = 633.94, p-value = 0.001838
## alternative hypothesis: true difference in means between group PM10 and
## group PM2.5 is not equal to 0
## 95 percent confidence interval:
## -32.904752 -7.526204
## sample estimates:
## mean in group PM10 mean in group PM2.5
## 139.6303 159.8457
```

7. ANOVA

```

anova_pollutant <- aov(pollutant_avg ~ pollutant_id, data = air)
summary(anova_pollutant)

##              Df   Sum Sq Mean Sq F value Pr(>F)
## pollutant_id    6 7768732 1294789   500.7 <2e-16 ***
## Residuals     2513 6497988    2586
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

top_cities <- names(sort(table(air$city), decreasing = TRUE)[1:10])
city_data <- subset(air, city %in% top_cities)

anova_city <- aov(pollutant_avg ~ city, data = city_data)
summary(anova_city)

##              Df   Sum Sq Mean Sq F value Pr(>F)
## city           9 1114358  123818   15.44 <2e-16 ***
## Residuals     786 6302920    8019
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

8. Regression Analysis

a. Simple Linear Regression

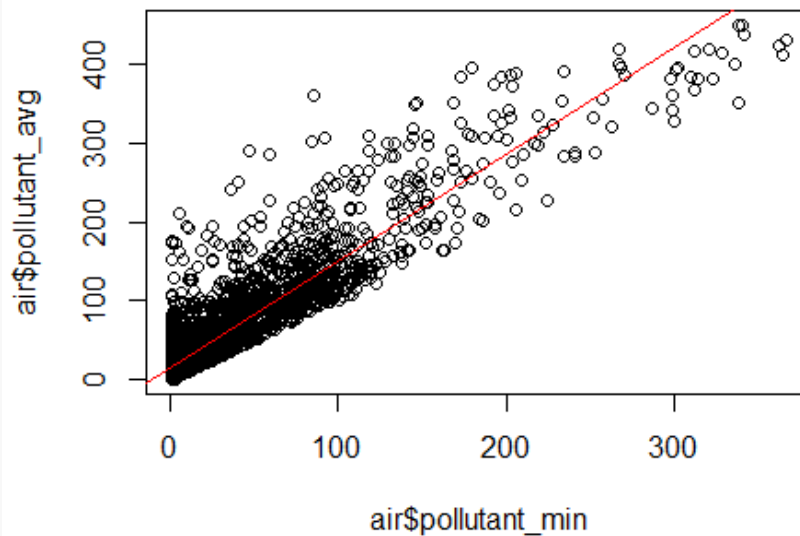
```

slr_model <- lm(pollutant_avg ~ pollutant_min, data = air)
summary(slr_model)

##
## Call:
## lm(formula = pollutant_avg ~ pollutant_min, data = air)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -121.944  -16.715  -11.364    9.621   230.855
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  15.31146    0.79510   19.26  <2e-16 ***
## pollutant_min  1.35098    0.01274  106.06  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.19 on 2518 degrees of freedom
## Multiple R-squared:  0.8171, Adjusted R-squared:  0.817
## F-statistic: 1.125e+04 on 1 and 2518 DF,  p-value: < 2.2e-16

```

```
plot(air$pollutant_min, air$pollutant_avg)
abline(slr_model, col = "red")
```



b. Multiple Linear Regression

```
mlr_model <- lm(pollutant_avg ~ latitude + longitude, data = air)
summary(mlr_model)
```

```
##
## Call:
## lm(formula = pollutant_avg ~ latitude + longitude, data = air)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -95.56  -48.79  -23.22   23.02  368.62
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  120.487     24.885    4.842 1.37e-06 ***
## latitude       2.355       0.277    8.502 < 2e-16 ***
## longitude     -1.392       0.316   -4.404 1.10e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 74.05 on 2517 degrees of freedom
## Multiple R-squared:  0.0327, Adjusted R-squared:  0.03193
## F-statistic: 42.55 on 2 and 2517 DF, p-value: < 2.2e-16
```

INFERENCE AND CONCLUSION

Inference

The statistical analysis of air quality data collected from various states and cities of India provides several meaningful insights regarding pollution patterns, pollutant distribution, and regional variations. The dataset consists of 2,520 observations covering multiple pollutants such as CO, NH₃, NO₂, OZONE, PM₁₀, PM_{2.5}, and SO₂. Using descriptive statistics, hypothesis testing, ANOVA, and regression analysis, the following inferences can be drawn.

1. Overall Pollution Level

The mean value of average pollutant concentration is 65.17, while the median is 39. The noticeable difference between the mean and median indicates a positively skewed distribution. This suggests that while many observations fall within moderate pollution levels, some extremely high values increase the overall average.

The standard deviation of 75.26 reflects high variability in pollution levels across regions. The maximum observed value of 449 further confirms the presence of extreme pollution events in certain locations. The interquartile range (IQR) of 76 indicates that the middle 50% of values are widely spread, demonstrating substantial dispersion in air quality conditions across the country.

From the histogram and boxplot, it can be inferred that the distribution is right-skewed with several outliers. These outliers represent cities or stations experiencing significantly higher pollution levels compared to the majority.

2. Distribution of Pollutant Types

The categorical distribution shows that pollutants are fairly evenly distributed across observations, with each pollutant contributing roughly 12–15% of total records. This balanced representation ensures that the dataset does not disproportionately favor one pollutant over others.

However, statistical comparisons reveal differences in average concentrations among pollutants. The ANOVA test for pollutant type produced a highly significant result ($p\text{-value} < 2e-16$), indicating that at least one pollutant has a significantly different mean concentration compared to others. This confirms that pollutant levels are not uniform across types and vary significantly.

3. Comparison Between PM2.5 and PM10

A Welch Two Sample t-test was conducted to compare the mean concentration of PM2.5 and PM10. The test resulted in a p-value of 0.001838, which is less than the 0.05 significance level. Therefore, the null hypothesis of equal means is rejected.

The mean concentration of PM2.5 (159.85) is higher than that of PM10 (139.63). This indicates that finer particulate matter (PM2.5) is present at significantly higher levels compared to coarser particulate matter (PM10). Since PM2.5 particles are more harmful due to their ability to penetrate deep into the lungs, this finding has important public health implications.

4. State-wise Pollution Ranking

The state-wise analysis reveals significant variation in pollution levels across regions. Delhi ranks highest with an average pollution level of 134, followed by Himachal Pradesh and Chandigarh. States such as Rajasthan, Haryana, and Maharashtra also show relatively high averages.

The high ranking of Delhi aligns with known environmental concerns in the National Capital Territory. Rapid urbanization, vehicular emissions, industrial activities, and seasonal factors likely contribute to this elevated level. The variation across states suggests that geographical, industrial, and demographic factors influence pollution intensity.

5. City-wise Variation

The ANOVA test conducted on major cities also resulted in a highly significant p-value ($< 2e-16$). This indicates that pollution levels differ significantly across cities. Urban centers with high population density and industrial activity likely experience greater pollution levels compared to smaller or less industrialized cities.

Thus, air quality management strategies should be region-specific rather than uniform nationwide.

6. Simple Linear Regression

The simple linear regression model examined the relationship between minimum pollutant level and average pollutant level. The regression equation shows a positive slope (1.35098), indicating that for every unit increase in minimum pollution, the average pollution increases by approximately 1.35 units.

The R-squared value of 0.8171 suggests that about 81.7% of the variation in average pollution can be explained by the minimum pollution value. This indicates a very strong linear relationship. The p-value ($< 2e-16$) confirms that the relationship is statistically significant.

Therefore, minimum pollution level is a strong predictor of average pollution level in this dataset.

7. Multiple Linear Regression

The multiple regression model evaluated the effect of geographical coordinates (latitude and longitude) on pollution levels. Although both variables are statistically significant, the R-squared value is only 0.0327. This means that only 3.27% of the variation in pollution is explained by location alone.

This suggests that while geography has some influence, other factors such as industrial density, traffic volume, weather conditions, and urban infrastructure play a much larger role in determining pollution levels.

Conclusion

The statistical analysis of air quality data from India provides clear evidence of substantial variation in pollution levels across pollutants, states, and cities. The descriptive statistics indicate a positively skewed distribution with high variability and extreme values. This suggests that while many regions experience moderate pollution, certain areas face severe air quality issues.

The hypothesis testing results confirm significant differences between PM_{2.5} and PM₁₀ concentrations, with PM_{2.5} being significantly higher. Given the serious health risks associated with PM_{2.5} exposure, this finding highlights the need for stricter monitoring and targeted control measures.

The ANOVA results demonstrate that pollutant levels differ significantly across pollutant types and across cities. This supports the idea that environmental conditions and emission sources vary widely across locations.

The regression analysis further strengthens the findings. The strong relationship between minimum and average pollution levels indicates internal consistency in pollution measurements.

However, the weak explanatory power of geographical variables suggests that pollution is influenced more by human activities than by mere spatial location.

In conclusion, the study successfully applies statistical tools to evaluate air quality patterns and identify significant differences and relationships within the dataset. The results emphasize the importance of region-specific environmental policies, continuous monitoring, and stricter emission control measures to improve air quality.

Future research may incorporate additional variables such as temperature, humidity, traffic density, and industrial output to build more comprehensive predictive models. Expanding the dataset across different time periods would also allow for trend analysis and seasonal comparisons.

Overall, the analysis demonstrates how statistical inference provides meaningful insights into real-world environmental problems and supports evidence-based decision-making for sustainable development.